

ST THOMAS CHARLOTTE AMALIE, Virgin Islands, U.S.

WMO: 994042

Lat: 18.3351N

Lon: 64.9197W

Elev: 16

StdP: 14.69

Time Zone: -4.00 (W04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	72.2	73.1	N/A	N/A	N/A	N/A	N/A	N/A	12.7	81.3	11.8	80.9	2.6	80	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.4	89.7	N/A	89.0	N/A	88.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.7	9.5	DB	70.1	91.1	0.8	1.0	69.5	91.8	69.0	92.4	68.6	93.0	68.0	93.7	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.4	78.6	78.5	78.6	79.9	81.5	83.3	83.9	84.2	83.8	83.3	81.6	79.9	(6)	
	DBStd	2.60	1.20	1.20	1.65	1.51	1.73	1.53	1.35	1.39	1.47	1.56	1.50	1.39	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	11476	887	799	886	898	976	998	1051	1060	1015	1031	949	926	(10)	
	CDD65	6001	422	379	421	448	511	548	586	595	565	566	499	461	(11)	
	CDH74	63464	3256	2865	3290	4111	5499	6529	7236	7451	6998	6753	5298	4178	(12)	
(13)	CDH80	19131	423	359	453	745	1471	2349	2856	3053	2775	2463	1449	735	(13)	
(14)	Wind	WSAvg	3.9	3.8	4.0	4.5	4.3	3.7	4.1	4.2	3.9	3.5	3.3	3.4	4.0	(14)
(15) Precipitation	PrecAvg	39.2	1.8	1.4	2.2	2.6	3.9	2.9	2.6	3.9	4.7	5.5	4.6	3.1	(15)	
	PrecMax	60.2	3.7	2.7	5.7	16.4	11.3	8.1	5.6	8.5	12.8	16.9	11.2	7.5	(16)	
	PrecMin	23.2	0.4	0.5	0.4	0.2	0.4	0.4	0.7	0.8	1.7	1.4	0.8	0.6	(17)	
	PrecStd	10.8	0.9	0.6	1.5	3.3	2.9	2.2	1.4	2.1	2.3	4.1	3.3	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.6	84.7	86.0	86.6	88.2	89.6	90.0	90.6	90.4	90.0	88.2	86.2	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	83.9	83.7	84.5	85.1	87.0	88.6	89.1	89.6	89.3	88.9	87.3	85.2	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	83.2	83.1	83.6	84.3	86.1	87.8	88.4	88.9	88.6	88.0	86.5	84.4	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
(25) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	82.5	82.3	82.6	83.6	85.3	87.1	87.8	88.1	87.8	87.2	85.6	83.6	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
(31) Mean Daily Temperature Range	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
	5% DB	MDBR	8.4	8.4	8.0	7.8	7.5	7.8	8.3	8.4	8.2	8.3	8.6	8.2	(35)	
		MCDBR	9.2	9.1	8.9	8.3	8.2	8.4	8.7	9.1	9.3	9.1	9.3	8.8	(36)	
(40) Clear-Sky Solar Irradiance	taub	taud	2.578	2.536	2.532	2.493	2.393	2.162	2.133	2.278	2.429	2.583	2.600	2.596	(40)	
		Ebn at Noon	290	292	294	285	271	251	249	261	273	285	284	285	(41)	
	Edh at Noon		30	33	34	36	39	48	50	43	37	31	29	29	(42)	
															(43)	
(44) All-Sky Solar Radiation	RadAvg		1513	1766	1970	2082	2007	2019	2047	2015	1889	1656	1457	1404	(44)	
	RadStd		40	38	74	119	134	119	102	74	120	117	101	62	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.41	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+142	+144

Nomenclature: See separate page